

Unit - 2

classmate

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$$(1) \frac{dy}{dx} + P(x)y = Q(x)$$

 $S P(x) dx$ I.F. = e

$$e^{\int P(x) dx} y = \int e^{\int P(x) dx} Q(x) dx + C$$

(2) Newton's Law of Cooling

$$\frac{dT}{dt} = -K(T - T_a)$$

(3) O.T. $\Rightarrow f(x) = 0$ Differentiate take $\left(\frac{dy}{dx}\right)$ then change it to $-\frac{dx}{dy}$ then integrate.

$$x^2 + y^2 = a^2$$

$$2x + 2y y' = 0$$

$$-\frac{x}{y} = \frac{dy}{dx} \Rightarrow \frac{-x}{y} = \frac{dx}{dy}$$

$$\log y = \log x + \log e$$

$$y = ex \leftarrow \text{O.T.}$$

- for Polar Curves

$$r \frac{d\theta}{dr} \Rightarrow \int -\frac{1}{r} \frac{dr}{d\theta}$$

$$r\theta = a$$

$$\frac{dr}{d\theta} = -\frac{a}{\theta^2}$$

$$-\frac{r\theta^2}{\theta^3} = r \frac{d\theta}{dr}$$

$$-\frac{r\theta^2}{\theta^3} = -\frac{1}{r} \frac{dr}{d\theta}$$

$$\frac{\theta^2}{2} = \log r$$

$$e^{\frac{\theta^2}{2}} c = r$$

Taylor Series Method

* $y' = (y^2 + x) \quad y(0) = 1$

T.S. $\rightarrow f(x) + x f'(x)$

$x_0 = 0 \quad y_0 = 1$

$y'' = 2yy' + 1 \quad y'(0,1) \Rightarrow 1$

$y''(0,1) = 2 + 1 = 3$

$y''' = 2(y')^2 + 2yy''$

$y'''(0,1) \Rightarrow 2 + 2 = 4$

$\therefore y = 1 + x + \frac{x^2}{2!} 3 + \frac{x^3}{3!} 4 + \dots$

* Runge-Kutta Method :-

$y_1 = y_0 + K$

$K_1 = h f(x_0, y_0)$

$K_2 = h f(x_0 + \frac{h}{2}, y_0 + \frac{K_1}{2})$

$K_3 = h f(x_0 + \frac{h}{2}, y_0 + \frac{K_2}{2})$

$K_4 = h f(x_0 + h, y_0 + K_3)$

$K = \frac{K_1 + 2K_2 + 2K_3 + K_4}{6}$

h = increment

$y' = (x+y)^{1/2} \quad y(0.4) = 0.41$

find $y(0.8)$ in 2 step

$y_1 = \dots \quad x_1 = 0.6$

Then find y_2 $\Rightarrow ? \quad x_2 = 0.8$

Ans

* Euler Method :-

$y(x_0 + h) = y(x_0) + h y'(x_0)$

$y(x_1) = y(x_0) + h y'(x_0)$

$y(x_2) = y(x_1) + h y'(x_1)$

n	x_n	y_n	$\frac{dy}{dx}$	y_{n+1}

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* Modified Euler Method

- Predictor step

$$y_1^{(0)} = y_0 + h f(x_0, y_0)$$

Corrector step

$$y_1^{(1)} = y_0 + \frac{h}{2} \left(f(x_0, y_0) + f(x_1, y_1^{(0)}) \right)$$

$$x_1 = h + x_0$$

$$y_2^{(2)} = y_0 + \frac{h}{2} \left(f(x_0, y_0) + f(x_1, y_1^{(1)}) \right)$$

$$x_1, y_1^{(2)} \leftarrow \text{Main Points}$$

$$y_2^{(0)} = y_1^{(2)} + h f(x_1, y_1^{(2)})$$

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

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$$\cosh x = \frac{e^x + e^{-x}}{2}$$

① Solve D.E. C Complementary Solution

$$y'' - 6y' + 9y = 0$$

$$(m^2 - 6m + 9) = 0$$

$$(m - 3)^2 = 0 \quad m = 3, 3$$

$$y = (C_1 + x C_2) e^{3x}$$

Complex $\Rightarrow$

$$\alpha \pm i\beta$$

$$e^{\alpha x} (C_1 \cos \beta x + C_2 \sin \beta x)$$

② Inverse Operator $\Leftrightarrow$ Particular Solution

Rule 1 -

$$e^{ax}$$

$$2^x \Rightarrow$$

$$e^{x \log(2)}$$

Change $\boxed{D \rightarrow a}$

$$\omega = \frac{-1 \pm \sqrt{3}i}{2}$$

$$(D^2 + 9D + 4)y = (1 + e^{-2x} + 2e^{-x})$$

$$y = \frac{1}{D^2 + 9D + 4} (1 + e^{-2x} + 2e^{-x})$$

$$\text{let } D \rightarrow 0 \quad D \rightarrow -2 \quad D \rightarrow -1$$

if one of D is Zero then

$$\frac{x}{(2D + 9)}$$

$$\text{or } \frac{x^2}{(2)}$$

(Differentiate)

Rule 2

$$\sin(ax+b) \text{ or } \cos(ax+b)$$

$$D^2 \rightarrow -a^2$$

Rule 3

$$(1+x)^{-1} \Rightarrow 1 - x + x^2 - x^3 + \dots$$

$$(1-x)^{-1} \Rightarrow 1 + x + x^2 + x^3 + \dots$$

$$(1+x)^2 \Rightarrow (1 - 2x + 3x^2 - 4x^3 + \dots)$$

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$$\frac{1}{D} (x^2 + 4) \quad \text{(Integration)}$$

$$\frac{1}{(x^2)} \quad \frac{x^3}{3} + 4x$$

$$(3 + D)$$

$$\frac{1}{3} \left(1 + \frac{D}{3}\right)^{-1} \cdot (x^2)$$

$$\frac{1}{3} \times \left(1 - \frac{D}{3} + \frac{D^2}{3}\right) x^2$$

Rule-4

$$\frac{1}{(D^2 + 2)} e^{ax} f(x)$$

$$e^{ax} \times \frac{1}{((D+a)^2 + 2)} f(x)$$

* Cauchy's

$$(2x-1)^2 y'' + (2x-1) y' - x^2 y'' + x y' + y = 2 \log(x)$$

Let $z = \log(x)$

$$(D'(D'-1)y + D'y + y) = 2 \log(x)$$

$$((D'(D'-1) + D' + 1)y = 2 \log(x)$$

Now same as rules,
find Complimentary & Particular

* Legendre

$$(ax+b)^2 y'' + (ax+b) y' + y = \log(ax)$$

$y = \log(ax+b)$

$$(a^2 D'(D'-1) + a D' + 1) y = \log x$$

* Method of Variation of Parameter

make constant as $V_1(x)$ & $V_2(x)$

$$y = V_1(x) \cos x + V_2(x) \sin x$$

$$2 \sin A \cos B = \sin(A+B) + \sin(A-B)$$

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Now, 2 eqⁿ

$$\begin{cases} V_1'(x) \cos x + V_2'(x) \sin x = 0 \\ V_1'(x) \sin x + V_2'(x) \cos x = 0 \end{cases}$$

Solve

get $V_1'(x) \& V_2'(x)$
integrate get $\int V_1'(x) \& V_2'(x)$

+ Constant

$$\begin{cases} V_1 = \int V_1'(x) + K_1 \\ V_2 = \int V_2'(x) + K_2 \end{cases}$$

Put in eqⁿ get solⁿ

* (Wronskian) $\frac{d^2 y}{dx^2} + 2y = \tan x \Rightarrow (X)$

$$y = \underbrace{V_1(x)}_{\text{constant}} \underbrace{\phi_1(x)}_{\text{constant}} + \underbrace{V_2(x)}_{\text{constant}} \underbrace{\phi_2(x)}_{\text{constant}}$$

$$W = \begin{vmatrix} \phi_1 & \phi_2 \\ \phi_1' & \phi_2' \end{vmatrix}$$

$$V_1'(x) = -\frac{\phi_2' x}{W} \quad V_2'(x) = \frac{\phi_1' x}{W}$$

X X

(Ch-4) Matrix

① Echelon form
Rank = ?

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix} \begin{matrix} \leftarrow \text{no. of non zero rows} \\ \leftarrow \rho(2) \\ \leftarrow \end{matrix}$$

→ This must be 0.

② $AX = 0$ if rank < n then non-trivial solⁿ exist
↓
(other than $(0, 0, 0)$)

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to solve always take other than pivot point as arbitrary points

$$\begin{matrix} & x & y & z & u \\ \begin{pmatrix} 2 & 1 & 0 & 1 \\ 0 & 0 & 4 & 4 \\ 0 & 0 & 0 & 0 \end{pmatrix} \end{matrix}$$

take y & u are arbitrary to solve least eqⁿ.

③ $AX = B$ (Gauss Elimination Method)

→ Right Augmented form $[A:B]$

→ Convert to Echelon

Rank A & Rank $[A:B]$

— If Rank $[A] = \text{Rank } [A:B] = r$
 $r = n = m$ then unique

— $r < n$, ∞ solⁿ

— Rank $[A] \neq \text{Rank } [A:B]$, NO solⁿ

④ Jacob's Method:— [Diagonally Dominant]

Take Diagonally Dominant elements & write
 x in terms of y & z
 y in terms of z & x
 z in terms of x & y

Now initial assumption as $x = y = z = 0$.
 Then repeat again

⑤ Gauss-Seidal Method

Directly use the update value u get.

$$\left. \begin{aligned} x_1 &= y(0) + z(0) \\ y_1 &= x_1 + z(0) \\ z_1 &= x_1 + y_1 \end{aligned} \right\}$$

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⑥ Eigen Values & Eigen Vectors

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = 0$$

$$\begin{vmatrix} (a_{11} - \lambda) & a_{12} \\ a_{21} & (a_{22} - \lambda) \end{vmatrix} = 0$$

$$(a_{11} - \lambda)(a_{22} - \lambda) - a_{12}a_{21} = 0$$

Solve to get $\lambda \leftarrow$ Eigen values

For λ_1

$$\begin{bmatrix} (a_{11} - \lambda_1) & a_{12} \\ a_{21} & (a_{22} - \lambda_1) \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Solve get eq

Then for λ_2 the same

⑦ Properties of Eigen Value & Eigen Vectors

- $\lambda_1 + \lambda_2 + \lambda_3 = a_{11} + a_{22} + a_{33} \leftarrow$ Diagonal elements

- $\boxed{\lambda_1 \lambda_2 \lambda_3 = |A|}$

- λ is A then $\frac{1}{\lambda}$ is A^{-1}

- λ is A then $\lambda^2 = A^2$
 then $\lambda^m = A^m$ }

- For diagonal matrix, upper Δ & lower Δ

$\lambda = a_{11}, a_{22}, a_{33}$

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⑧ Repeated Eigen Value :-

- For 3×3 has some Eigen Value Rank = 1
 $\therefore$ 2 Arbitrary Value

$$\begin{aligned} \text{Then Case I} &\Rightarrow K_1 = 0 \quad K_2 = 1 \\ \text{Case II} &\Rightarrow K_1 = 1 \quad K_2 = 0 \end{aligned}$$

⑨ Rayleigh's Power Method:-

$$A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$$

$$y' = A y^{(0)}$$

$$y^0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$y^2 = A y^1$$

$$y^1 = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$$

$$y_0 = 3 \begin{bmatrix} 1 \\ 1/5 \end{bmatrix}$$

⑩ Diagonalization of Matrix:-

$$D = P^{-1} A P$$

$$A = P D P^{-1}$$

$$A P = P D$$

$$A = \begin{bmatrix} 5-\lambda & 4 \\ 1 & 2-\lambda \end{bmatrix} = 0$$

$$\lambda = 1, 6$$

$$\lambda = 1 \Rightarrow \begin{bmatrix} 4 & 4 \\ -1 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\lambda = 6 \Rightarrow \begin{bmatrix} -1 & 4 \\ 1 & -4 \end{bmatrix} \rightarrow \begin{bmatrix} 4x_2 \\ x_2 \end{bmatrix} \rightarrow \begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

$$A^n = P D^n P^{-1} \quad P = \begin{bmatrix} -1 & 4 \\ 1 & 1 \end{bmatrix}$$

$$11) \quad y' = A y \rightarrow A - \lambda I \rightarrow \lambda$$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = c_1 x_1 e^{-3t} + c_2 \begin{bmatrix} 4 \\ 1 \end{bmatrix} e^{6t}$$

2 eq^s solve to get y_1, y_2